Linux Computer Literacy Report

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1 File Manager - Directories

One of the most important aspects of using a file manager is understanding how to navigate and manage directories effectively. This includes creating, renaming, moving, and deleting directories, as well as understanding the hierarchical structure of the file system.

A directory is a container used to organize files on a computer. It can contain files as well as other directories, which are often referred to as subdirectories.

2 Python & pip

2.1 Installing Python

Fist you need a compiler like VScodium. This can be downloaded and installed from its official website or through your system's package manager and the terminal. you can run the command sudo apt install python3 to install Python on your system. Make sure to verify the installation by running python3 –version in the terminal.

2.2 Python Files

PY, IPYNB, and other Python-related files are commonly used in Python development. PY files are standard Python scripts, while IPYNB files are Jupyter Notebook files that allow you to create and share documents containing live code, equations, visualizations, and narrative text. It is important to understand the differences between these file types and how to work with them effectively in your Python projects.

2.3 Installing pip

To install pip, you can use the following command:

```
python -m ensurepip --upgrade
```

Note: you may have a different version of python installed just replace `python` with the appropriate version, e.g., `python3`.

2.4 Installing Packages

To install packages using pip, you can use the following command:

```
pip install "package_name"
```

If a Virtual Environment is not set up this will install the package to the site-packages directory of your Python environment.

3 Venv

3.1 Creating a Virtual Environment

A virtual environment is an isolated Python environment that allows you to manage dependencies for different projects separately. You can create a virtual environment using the following command:

```
python -m venv myenv
```

3.2 Activating a Virtual Environment

To activate the virtual environment, use the following command:

```
source myenv/bin/activate  # On Unix or MacOS
myenv\Scripts\activate     # On Windows
```

4 LaTeX In VScodium

LaTeX is not important for first or second-year students, although some people do use it, you will not get any extra marks, and if you dont know what your doing it is not better than using just a regular text editor text editor. Side note: microsoft word can also use LaTeX but you will have to figure that out yourselfs.

VScodium is a lightweight, open-source code editor based on Visual Studio Code. Some people may choose to use OverLeaf to compile and preview their LaTeX documents online, however this is slow and may require you to pay for a subscription to compile you work. In my opinion, using VScodium for LaTeX is a much better alternative, it means you can compile and preview your documents locally without relying on an internet connection or a paid subscription.

The extension i use is called `LaTeX Workshop`. I havent yet Tried anything else as this works just fine. Make sure to install this extension from the VScodium marketplace to enable LaTeX support, including compilation, preview, and syntax highlighting. If you are reading this after the talk, you can follow the instructions in the appendix to set up VScodium for LaTeX.

5 BASH

BASH (Bourne Again SHell) is a command-line shell used for interacting with the operating system. It allows you to execute commands, run scripts, and automate tasks efficiently. the command "- -help" after any command to get more information about its usage. or the command "man" followed by the command name to access the manual. For example:

5.1 Basic BASH Commands

```
# List files and directories
ls

# Change directory
cd /path/to/directory

# Create a new directory
mkdir new_directory

# Remove a file
rm file_name

# Remove a directory and its contents
rm -r directory_name

# Display the current directory
pwd

# stop running processes in the terminal
```

```
Ctrl+C

# copy and paste changes picking up a shift
Cntrl+Shift+V  # Paste in terminal
Cntrl+Shift+C  # Copy in terminal

# stop the terminal
exit

# clear the terminal
clear

# run as administrator
sudo command_here

# search for a file or directory
find path -name file_name

% open vscodium
codium . # Open the current directory in VScodium
codium file_name  # Open a specific file in VScodium
```

6 GIT

GIT is a version control system that allows you to track changes in your code and collaborate with others. it is usefull for managing code changes and collaborating with others.

6.1 Git Commands

```
echo "# Example_Project" >> README.md
git init
# Initialize a new Git repository

git add README.md
# Stage the README file for commit

git commit -m "first commit"
# Commit the staged changes with a message

git branch -M main
# Rename the default branch to main

git remote add origin git@github.com:UserName/Example_Project.git
# Add the remote repository

git push -u origin main
# Push the changes to the remote repository

git remote add origin git@github.com:UserName/Example_Project.git
# Add the remote repository
```

```
git branch -M main
# Rename the default branch to main
git push -u origin main
# Push the changes to the remote repository
```

In your degree you will use git to manage your code, track changes, and collaborate with your peers on various projects.